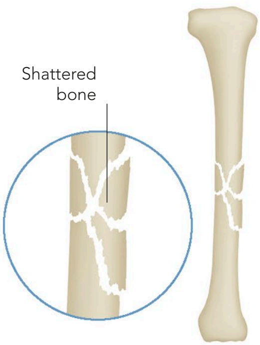
BONE AND JOINT DISORDERS

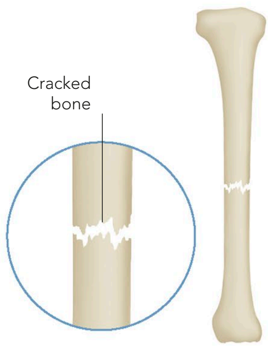
BONES AND JOINTS ARE VULNERABLE TO INJURIES SUCH AS FRACTURES AND, DUE TO CONSTANT WEAR, TO DISORDERS SUCH AS OSTEOARTHRITIS. BONES MAY BE WEAKENED BY OSTEOPOROSIS, AND JOINTS MAY BE AFFECTED BY INFLAMMATORY CONDITIONS SUCH AS RHEUMATOID ARTHRITIS.

FRACTURE

Fractures may be caused by a sudden impact, by compression, or by repeated stress. A displaced fracture occurs when the broken surfaces of bone are forced from their normal positions. There are various types of displaced fracture, depending on the angle and strength of the blow. A compression fracture occurs when spongy bone, such as in the vertebrae, is crushed. Stress fractures are caused by prolonged or repeated force straining the bone; they occur in long-distance runners and in the elderly, in whom minor stress, such as coughing, may cause a fracture. Nutritional deficiencies or certain chronic diseases such as osteoporosis, which can weaken bone, may increase the likelihood of fractures. If a broken bone remains beneath the skin, the fracture is described as closed or simple, and there is a low risk of infection. If the ends of the fractured bone project out through the skin, the injury is described as open or compound, and there is a danger of dirt entering the bone tissue and causing microbial contamination.

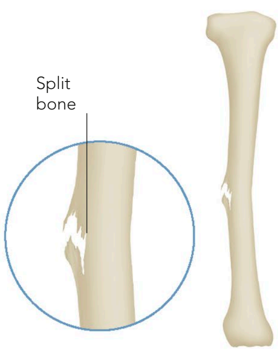


COMMINUTED FRACTURE
A direct impact can shatter a bone into several fragments or pieces. This type of fracture is likely to occur during a road traffic accident.



TRANSVERSE FRACTURE
A powerful force may cause a break across the bone width. The injury is usually stable; the broken surfaces are unlikely to move.

GREENSTICK FRACTURE
If a long bone bends under force, a crack may occur on one side. This type of fracture is common in children, whose bones are flexible.



SPIRAL FRACTURE
A sharp, twisting force may break a bone diagonally across the shaft. The jagged ends may be difficult to reposition.